DATA ANALYTICS & BUSINESS INTELLIGENCE

E-Commerce Sales Analysis Dashboard

An interactive Power BI dashboard built to turn a year of raw sales data into clear, decision-ready insight.

Project Report
Prepared July 2026
Tools: Power BI Desktop · Power Query · DAX · Data Modeling

1. Executive Summary

Two hundred thousand orders, one dashboard, and a much clearer picture of where the revenue comes from.

This report walks through an e-commerce sales analysis dashboard built in Power BI, covering a full year of transaction records across ten Indian cities, five product categories, and a base of ten thousand customers. Rather than leaving the numbers buried in transactional tables, the dashboard organizes them into KPI cards, charts, and interactive filters so that a category manager, marketing lead, or business owner can ask a question — "which brand is really driving revenue?" or "is our average order value actually growing?" — and get an answer in seconds, not hours.

The headline numbers tell a clear story: 636M in total revenue, 200K orders, and 493K units sold, each up roughly 16% year-over-year — while the customer base held flat at 10K and average order value slipped marginally. Clothing and Electronics lead the category mix, Zenith is the runaway top brand, and revenue is spread evenly across major metros rather than concentrated in any one city. The sections below unpack how the dashboard reaches these conclusions and what they suggest for action.

2. Project Overview & Objectives

The E-Commerce Sales Analysis Dashboard was built to make a large, multi-dimensional sales dataset usable at a glance. Raw order-level rows rarely tell a story on their own — it takes structure, visualization, and the ability to slice the data from different angles before patterns start to emerge. That's exactly what this dashboard does: it takes revenue, order, customer, and product-level records and reorganizes them into a format where trends across categories, brands, cities, and customer segments become visible immediately.

The project was guided by a few clear goals:

- Give a real-time, at-a-glance view of overall business performance
- Track Year-over-Year (YoY) growth on every headline KPI, not just raw totals
- Understand who the customer actually is — age, gender, and geography — alongside how much they spend
- Identify the categories, brands, and products carrying the most revenue
- Make city-level performance visible, since a national business is really a collection of local ones
- Let users explore freely through slicers and page navigation, without ever touching or risking the underlying data

3. Tools & Dashboard Features

Everything here was built natively in Power BI, using Power Query for data cleaning and shaping, DAX for calculated measures and YoY comparisons, and native data modeling to connect revenue, customer, and product tables into a single analytical model. Interactive filters, KPI cards, and a consistent Microsoft / Fortune 500–inspired visual style round out the experience.

Interactive Filters & Navigation

Each of the three dashboard pages carries its own slicer panel — Year, Category, City, Brand, or Gender, depending on the page — and a left-hand navigation menu lets the user move between Executive Overview, Customer Analysis, and Product Performance without losing their filter context.

KPI Cards

Every page opens with a row of KPI cards showing the metric alongside its Year-over-Year change, so the first question anyone asks — "how are we doing compared to last year?" — is answered before a single chart is read.

Visualizations

Across the three views, the dashboard combines line charts (monthly revenue trend), donut and treemap charts (category and city mix), horizontal bar charts (brand and product rankings), scatter plots (quantity vs. category), and a geographic map (customer distribution) — each chosen to answer a specific business question rather than for visual variety alone.

4. Walkthrough of the Dashboard Views

The clearest way to understand what this dashboard can do is to look at each page in turn — so the following pages walk through the three core views, what each one shows, and what it reveals about the business.

4.1 Executive Overview — Overall Business Performance

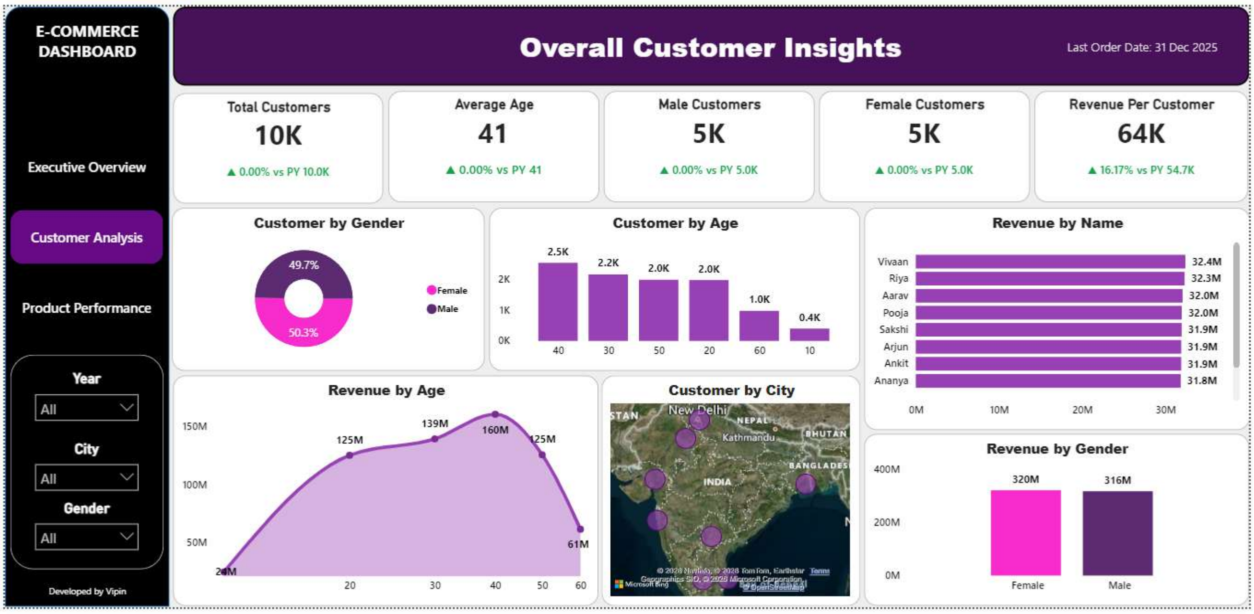
The business-performance master view — revenue, orders, and category/brand/city mix, all at once.

636M Total Revenue	**200K** Total Orders	**10K** Total Customers
493K Units Sold	**3.18K** Avg. Order Value	**+16.17%** Revenue Growth (YoY)

This is the dashboard at rest, showing the full year of data. Revenue, orders, and units sold all grew by roughly 16% against the prior year — three numbers moving in near-perfect lockstep, which is the clearest sign that this year's growth came from more transactions rather than pricier ones. Average Order Value tells the other half of the story: it actually dipped slightly to 3.18K, confirming that customers aren't spending more per basket — they're simply shopping more often.

Looking city by city, **Hyderabad** and Bangalore sit at the top with 77M each, just ahead of Chennai and Ahmedabad (75–76M); the spread across all ten metros is narrow, pointing to a mature, evenly distributed footprint rather than dependence on any single city. The category mix is similarly balanced — Electronics leads at 23% of revenue, with Clothing, Groceries, Accessories, and Home Decor all within a few points of each other. Month to month, revenue holds a tight band between roughly 49M and 55M, with a dip in February and a broadly flat trend for the rest of the year — a business with limited seasonality. Finally, **Zenith** is the clear brand leader at 78M, and **Clothing** is the single highest-revenue product line at 126M.

4.2 Customer Analysis — Overall Customer Insights



Isolating the customer base to see who is actually driving the revenue.

10K	41	64K
Total Customers	Average Age	Revenue / Customer

Narrowing the focus to customers rather than transactions changes the story in one important way: the customer base itself did not grow at all — it held flat at 10,000 — while revenue per customer rose 16.17% to 64K. That single figure confirms what the Executive Overview hinted at: the year's growth was earned from the existing base, not from acquiring new shoppers.

The base is almost perfectly split by gender (50.3% male vs. 49.7% female), and skews toward middle age — the 40-year-old bracket is both the largest customer segment and the single highest revenue-generating age group, peaking at 160M before tapering off at the older and younger ends of the curve. **Revenue by name** is tightly clustered at the top, with the leading eight customers each contributing within a narrow 31.8M–32.4M band — a sign that high-value revenue is broadly distributed rather than concentrated in a handful of outsized accounts. The city map confirms the same pan-India spread seen on the Executive Overview page.

4.3 Product Performance — Overall Product Performance

Zooming into the catalogue to see which categories, brands, and SKUs are actually earning their shelf space.

500 Total Products	**50M** Products Sold	**1.29K** Avg. Selling Price

The catalogue holds a stable 500 active products, with 50M total units sold across the year, up 16.23% year-over-year — while average selling price slipped a marginal 0.01%, once again pointing to volume as the growth driver rather than price. **Electronics** leads category revenue at 146M, narrowly ahead of Accessories (130M) and Clothing (126M), while **Zenith** repeats as the top brand at 78M, consistent with every other view in this report.

The quantity-vs-price scatter plot shows a steady, near-linear relationship across the catalogue, suggesting consistent unit economics rather than sharp outliers. In the products-by-category movement chart, all five categories show more products improving than declining, led by Electronics with 115 products trending up. Read alongside the top-products chart — where Clothing (9.8M units), Electronics (8.0M), and Groceries (7.5M) dominate volume while Tablet, Headphones, and Bluetooth Speaker remain under half a million units each — the picture is one of five well-established core categories and a small tail of low-volume accessory items that may need repricing, bundling, or a portfolio review.

5. Chart Reference Guide

For quick reference, here's what each visualization on the dashboard is built to answer:

Chart	Page	What it answers
Revenue by Month	Executive Overview	Whether revenue is seasonal or holds steady across the year
Revenue by Category / Brand	Executive Overview	Which categories and brands generate the most revenue
Revenue by City	Executive Overview	How revenue is distributed geographically across metros
Top 10 Products	Executive Overview	Which individual products contribute the most to revenue
Customer by Gender / Age	Customer Analysis	The demographic makeup of the customer base
Revenue by Age	Customer Analysis	Which age group actually drives the most spending
Revenue by Name	Customer Analysis	Whether top-line revenue is concentrated or broadly spread across customers
Customer by City (map)	Customer Analysis	Where the customer base is physically located
Revenue / Quantity by Category	Product Performance	How category-level revenue compares to units sold
Products by Category (movement)	Product Performance	Whether more products are trending up or down within each category

6. Key Insights & Recommendations

Pulling everything together, a few findings come through clearly and consistently across the different dashboard views:

- Growth is volume-led, not price- or customer-led: revenue, orders, and units sold all grew ~16%, while Average Order Value was flat-to-down and the customer base did not expand.
- Zenith is the dashboard's most consistent top performer, leading brand revenue on both the Executive Overview and Product Performance pages regardless of how the data is sliced.
- Clothing and Electronics are the twin growth engines — Clothing leads individual product revenue while Electronics leads category revenue, making them the natural priority for inventory planning and marketing spend.
- The customer base skews toward middle age (peak at 40) and is gender-balanced, meaning campaigns targeting the 35–50 age band are likely to have the highest return.
- Revenue is geographically well diversified across ten metros, reducing city-concentration risk but also meaning there is no single city-level lever for outsized growth.
- A small set of accessory SKUs sell at a fraction of core-category volumes and are worth a repricing, bundling, or delisting review.

Based on these patterns, a few practical next steps are worth considering: prioritize customer acquisition campaigns to complement the current base-expansion-led growth model; protect and expand the Clothing and Electronics ranges given their outsized contribution; and review the long tail of low-volume accessory products for repricing or bundling opportunities before the next planning cycle.

7. Closing Thoughts

What this project really demonstrates is that Power BI, used well, can turn a flat set of transactional tables into something a business can actually interrogate — filter it, compare it, and trust the answer that comes back. With interactive slicers, YoY-aware KPI cards, and a consistent visual language across three purpose-built pages, raw order data becomes a genuine decision-support tool rather than a static report.

This dashboard is offered as a working example of that idea — practical, interactive, and built to support real conversations about revenue, customers, and product strategy, not just report on the past.